

Program overview

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Year 2023/2024
Organization Aerospace Engineering
Education Minors Aerospace Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Minors LR 2023							
LR-MI-176-23 Offshore Wind Energy							
AE3512-20	Asset Management	5					
AE3513	Integratieopdracht	6					
AE3515	Basics of Aeroacoustics for Wind Energy	3					
AE3516A	Fundamentals of Wind Energy I	3					
AE3516B	Fundamentals of Wind Energy II	3					
CT3101	Project Management Basics	5					
TBM024B	Introduction to Energy Systems	5					
LR-MI-228-212 Airport Development							
LR-MI-228-23 Airport Development							
AE1110-I	Introduction to Aerospace Engineering I	5					
AE3501-19	Air Transportation	3					
AE3502-14	Airport Planning, Design and Operations	4					
AE3503	Strategic Planning for Airport Systems	6					
CT3080LR	Landside accessibility of Airports	6					
IO3818	Designing an Airport Ecosystem	6					
TB241TB	Logistics 2	5					
LR-MI-232-23 Space Missions							
Standard programme							
AE3522UL	Fundamentals of Astronomy	4					
AE3523	Planetary Sciences	3					
AE3524	Spacecraft Design and Astrodynamics	5					
AE3529	Space Instrumentation	5					
AE3538	Space Mission: Engineering Case Study	5					
AE3539UL	Space Mission: Scientific Case Study	5					
CT3533	Earth Observation	3					
Programme for Astronomy students (Leiden University)							
AE3523	Planetary Sciences	3					
AE3524	Spacecraft Design and Astrodynamics	5					
AE3524E	Spacecraft Design and Astrodynamics Extended	4					
AE3529	Space Instrumentation	5					
AE3538	Space Mission: Engineering Case Study	5					
AE3539UL	Space Mission: Scientific Case Study	5					
CT3533	Earth Observation	3					
Programme for EUR students							
AE3522UL	Fundamentals of Astronomy	4					
AE3523	Planetary Sciences	3					
AE3524	Spacecraft Design and Astrodynamics	5					
CT3533	Earth Observation	3					

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LR-MI-176-23 Offshore Wind Energy	
Minor Coordinator	Dr.ir. A.C. Viré

AE3512-20	Asset Management	5
Responsible Instructor	Dr. D. Zappalá	
Instructor	Dr. J.J.E. Teuwen	
Instructor	Dr. R.M. Groves	
Instructor	Prof.dr. S.J. Watson	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic coding knowledge (Python/Matlab or similar programming languages).	
Course Contents	Asset management concerns all activities related to the primary company values relative to asset ownership and use. It is most visible in execution of the strategic plan which an organization must have in place to optimally manage its assets for realizing its core values. Optimal asset management incorporates addressing the balance between performance, risk and costs over the entire asset lifecycle. For many asset-intensive companies, asset management is the primary business process. In recent years, various standards have been developed for asset management which are now being implemented in industry. An important element in these standards is that the optimal balance between costs, performance and risks are addressed at the strategic level (from a systems perspective) and covers the entire portfolio of assets. To reach the aforementioned optimal balance, asset-intensive companies require significant volumes and variety of data (e.g. asset condition data, contractual data, operational data, etc.). This course provides basic systems-oriented asset management concepts and tools, and introduces students to the principles, challenges and solutions faced by the wind industry. Effective asset management is crucial for optimizing the lifecycle cost of offshore wind energy which is a rapidly developing industry sector. The course combines state-of-the art theory with concrete examples and practical applications to wind energy assets. During this course students have the unique opportunity to directly hear from and engage with world-leading industry and academic experts on wind farm asset management. They are introduced to the current practice and innovation in industry as well as to the state-of-the-art and latest developments of the research in the field. The study material of the course includes a selection of research articles, which is regularly updated as novel research is generated. Hands-on assignments give students the opportunity to apply what they learn during the lectures to real wind farm management-related problems.	
Study Goals	In this course, you will learn the foundation of the theory of modern systems-oriented asset management and how to extend and apply this knowledge to current and future offshore wind farms in order to optimise their life-cycle costs. By the end of this course, you should be able to: L01: Describe the primary asset management concepts, tools and policies and their applications to offshore wind farms. L02: Apply the key concepts (RAMS) governing the management of wind assets. L03: Analyse the most cost-effective and intelligent maintenance and logistics strategies and policies for offshore wind farms. L04: Evaluate state-of-the-art condition monitoring data acquisition and analysis tools of wind assets and ways for their future improvements. L05: Formulate strategies to mitigate the main issues (from design to end-of life) affecting the life-cycle costs of offshore wind turbine blades.	
Education Method	The course consists of on-campus lectures. For a number of topics, guest lecturers from academia and industry are invited.	
Assessment	The final assessment consists of a written closed-book exam with open questions. Additionally, a bonus point (maximum of 0.5 out of 10 points) can be earned by performing the hands-on formative assignments available on Brightspace during the course. The assignments may involve small coding problems, open questions, or mathematical exercises. The bonus point is only valid for the regular exam of that academic year. not for the resit.	

AE3513	Integratieopdracht	6
Responsible Instructor	Dr.ir. D.A.M. De Tavernier	
Instructor	Dr.ir. A.C. Viré	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The assignment consists in answering key questions in the future development of offshore wind energy. The questions can relate to technical challenges, as well as environmental and management aspects. As far as possible, the assignment is done in collaboration with a company or a governmental body.	
Study Goals	At the end of the assignment, the student should be able to: - integrate knowledge from the different courses and sources, - reflect on challenges and trends for the future of offshore wind power, - report the work in a coherent written document and oral presentation of high academic standards, - work in a team to answer questions of industrial interest.	
Education Method	Independent teamwork.	
Assessment	Written report and online oral presentation.	

AE3515	Basics of Aeroacoustics for Wind Energy	3
Responsible Instructor	Dr. D. Ragni	
Instructor	L.T. Lima Pereira	
Responsible for assignments	Dr. D. Ragni	
Co-responsible for assignments	L.T. Lima Pereira	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching the world of sustainable energies. In wind-energy, new wind farms are designed with extremely large rotors. Off-shore wind-farms are additionally built relatively closer to the coast line and to populated areas, in order to benefit from those locations where the winds are relatively stronger. The aim of this course is to provide the students with a fundamental knowledge about noise sources with full focus and attention on the problem of wind-turbine noise. The course is structured in three main areas: 1) Introduction to noise sources and sound: This module introduces the basic knowledge about sound, sound pressure levels, sound power level and wave propagation. 2) Airfoil self-noise: In this module the students will study the fundamental sources of noise in airfoils operating at high-Reynolds number, with extension to blade noise for wind-turbine applications. 3) Wind-turbine noise The third area of the course deals with the theory behind wind-turbine noise. Experimental and numerical applications will be taken into account with an industrial view to the problem and to the most interesting noise-mitigation strategies.	
Study Goals	The course is meant to give the students a basic knowledge of wind-turbine noise for on/off shore applications. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Know what sound is and how noise is generated - Describe the most important sources of noise for airfoils and how to translate that to a wind-turbine blade - Analyze the most important sources of noise for wind-turbine rotors and their consequences for on/off shore applications	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Wind-Turbine Noise, Authors: Wagner, Siegfried, Bareiß, Rainer, Guidati, Gianfranco - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of a written exam with multiple choice and open questions.	

AE3516A	Fundamentals of Wind Energy I	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr. D.A. von Terzi	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course introduces the topic of offshore wind energy by going through the main steps that a wind farm developer needs to achieve in order to build an offshore wind farm. The main topics cover in the course are: - Historical developments in wind energy - Wind energy harvesting and Betz limit - Offshore wind and wave conditions - Estimation of the energy yield - Design considerations for wind turbine farms - Economic aspects - Planning issues - Emerging concepts such as floating wind turbines	
Study Goals	At the end of this course, the students will be able to: - Explain the basic principles of wind energy conversion. - Identify the steps required when developing offshore wind farms. - Differentiate between the challenges and opportunities of offshore and onshore wind energy harvesting. - Use existing tools for estimating the power produced by an offshore wind farm and the effect of wakes. - Integrate knowledge from various fields of engineering related to the development of offshore wind parks. - Reflect on the current and future trends in the field of offshore wind energy.	
Education Method	The teaching method uses the principle of flipped classroom. This means that, every week, the students have to perform online assignments (e.g. watch a lecture, read a text, etc) prior to coming to class. In class, additional lecture materials, exercises and feedback are given to the students.	
Assessment	The final assessment is a written exam. Additional points are given for performing the online activities prior to class.	

AE3516B	Fundamentals of Wind Energy II	3
Responsible Instructor	Dr. D.A. von Terzi	
Instructor	Dr.ir. A.C. Viré	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course introduces the topic of offshore wind energy production from the perspective of a wind turbine Original Equipment Manufacturer (OEM). The main questions covered in the course are: - Why harvest offshore wind energy? - What are key components of a wind turbine? - What are objectives and steps in the wind turbine design process? - What are key technical, economic and social acceptance related challenges and opportunities for offshore wind energy? - What are current trends and emerging technologies in offshore wind energy?	
Study Goals	At the end of this course, in the context of offshore wind, the students will be able to: - Identify different stakeholders and reflect on differences in their interests. - Explain the basic components of a wind turbine. - Explain the objectives of the wind turbine design and name the steps of the process. - Identify technical, financial and social acceptance related challenges and opportunities. - Reflect on current trends and emerging technologies.	
Education Method	The teaching method uses a standard classroom approach with activation elements. This means that, in class, exercises and feedback are given to or collected from the students. Supplemental material for further study is provided after each class.	
Assessment	The final assessment is a written exam.	

CT3101	Project Management Basics	5
Responsible Instructor	M.G.C. Bosch-Rekvelde	
Instructor	S.E.M.A. Elmohr	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is an important cornerstone of the minor Project Management: from Nano to Mega. In this course the general PM principles will be taught such as: iron triangle, trade-offs, scoping, scheduling, risk management, monitoring and control, project start-up, project close-out. Also the more strategic context of a project in an organization is discussed including portfolio management and the business case of a project. Throughout, the project life-cycle acts as a basis: A. Conception B. Definition C. Execution, contracting, implementation D. Handover Trade-offs between time, cost, quality, and scope are extensively discussed, after discussing the different axes in detail: -SCOPE axis - setting the project boundaries, WBS; -TIME axis - planning & scheduling, network scheduling, critical path method (CPM), Program Evaluation and Review Technique (PERT), resource constraints; -COST axis - estimating, project cost accounting, scheduling and forecasting for costs; -QUALITY axis - risk, TQM, six sigma, other techniques. Attention is paid to both front-end development and project execution. During the project execution the execution team utilizes all the schedules, procedures and templates that were prepared and anticipated during prior phases. Unanticipated events and situations will inevitably be encountered, and the Project Manager and Project Team will have to deal with it (change management). Overall: People are key in successful project management!	
Study Goals	The course is aimed at getting to know the basics of project management and being able to apply the obtained knowledge on real life cases. More specifically, after following this course the student is able to 1. Explain the difference between projects, programs and portfolios. 2. Explain how value is developed in projects throughout the project life cycle (PLC). 3. Select and reflect upon an appropriate organisation structure for a project. 4. Construct a project planning using MS Project /PertMaster. 5. Create a risk register based upon a qualitative risk analysis. 6. Explain the role of change management techniques in project management.	
Education Method	This course will be mainly based on lectures (/guest lectures) with smaller assessed weekly assignments for elaborating on the various topics. We will also use study materials from the MOOC Management of Engineering Projects: Preparing for success!	
Course Relations	This course is part of the minor Project Management: from Nano to Mega (CT-MI-174). Also students of the minor Offshore Wind Energy follow this course. Non-minor students might participate as well, but always should contact the lecturer upfront.	
Assessment	The assessment consists of - Written exam (half knowledge questions, half questions related to a casus) - Assignments (obligatory), further specified during lecture 1, which all should be completed before the final grade of the exam can be obtained. The course grade is based on the exam grade (but only when all assignments have been completed sufficiently).	
Tags	Business Project planning / management	
Expected prior Knowledge	No specific prior knowledge required.	
Academic Skills	This course will develop your writing and reflecting skills. And of course your project management skills!	
Literature & Study Materials	Management of Engineering Projects - People Are key. Hans L.M. Bakker and Jaap P. de Kleijn, editors. NAP - The Process Industry Competence Network. ISBN/EAN: 978-90-812162-0-3 This book can be ordered (student-price) via NAP or via the minor-staff (will be communicated in lecture 1).	
Judgement	The weekly assignments will be graded with fail - sufficient - good. All weekly assignments need to be at least sufficient: for a "fail", rework has to be done. The course grade is based on the exam grade (but only when all assignments have been completed sufficiently). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	No materials are permitted during the Exam, except Dutch-English (or vice versa) dictionaries. Also a simple calculator can be used.	
Colleggerama	Yes	

TBM024B	Introduction to Energy Systems	5
Responsible Instructor	Dr.ir. M.D.M. Pérez-Fortes	
Module Manager	Dr.ir. M.D.M. Pérez-Fortes	
Assistant	I. Stepchuk	
Assistant	T.J. Wiltink	
Responsible for assignments	Dr.ir. M.D.M. Pérez-Fortes	
Co-responsible for assignments	Prof.dr.ir. C.A. Ramirez Ramirez	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This is a core course to the Offshore Wind Energy Minor from AE. It is also offered as an elective course.	
Expected prior knowledge	Basic math, physics, and chemistry/technology knowledge are required. Perform basic mass and energy balances.	
Course Contents	Energy systems embrace multiple technologies, sectors, and perspectives. In this course, you will understand the multiscale (i.e. different levels of aggregation, i.e. equipment, household/industry, energy chain, sector, region, country, international, world) and multiperspective (i.e. technical, environmental, economic, social, and governance) nature of the analysis of energy systems and you will be able to (i) apply methodologies to estimate technical, environmental and economic performance and (ii) recommend options to improve performance at different levels of aggregation. The course provides the foundation for the understanding of the main elements that characterise energy systems. Special attention is given to global developments and past, current, and future trends. This includes the current transition from a fossil-fuel based system to a climate-neutral energy system and related potentials and scenarios. Energy systems analysis is a field where good argumentation skills are essential, so in addition to calculations, there is a significant focus on critical thinking, discussion, and reflection.	
Study Goals	By the end of this course, you should be able to: - Discuss the characteristics of an energy system at different levels of aggregation. - Calculate technical, environmental, and economic indicators at different levels of aggregation. - Estimate system performance based on calculated indicators and contextual information. - Compare energy policies and market issues that would impact a selected system at a specific level of aggregation.	
Education Method	Lectures and self-study/homework. The course counts with guest lectures from experts working in the different aspects of energy systems. The book Introduction to Energy Analysis by Kornelis Blok and Evert Nieuwlaar, third edition, is used to complement the content of the lectures and for homework problems. Each week there are two interactive lecture classes (2 hours each) and a problem-solving session (1 hour). There will be room for Q&A. To take the most of this course you should participate in class and do the homework.	
Literature and Study Materials	Introduction to Energy Analysis by Kornelis Blok and Evert Nieuwlaar, third edition. Lecture slides (Brightspace) and past edition recorded lectures (Collegerama).	
Assessment	The assessment consists of an individual assignment (report) and a written open-book exam. The individual assignment has a weight of 40%. The exam takes place at the end of the course (week 10 - 9 Nov, and lasts for 3 hours) and has a weight of 60%. To pass the course, individual assignment and exam must be graded with at least a 5.0 (minimum combined grade of 5.75). When the grade is not sufficient (individual grades below 5 or grade combined below 5.75), the resit for both summative assignments takes place in January (week 2.10). The results of both summative assignments are transferable to the next academic year (when no change is made in the evaluation method). Upon request, you will have the right to inspect your assessed work for a period of at least 20 working days after the announcement of the results. Review dates will be proposed, and you will have the opportunity to register for one specific time slot.	
Permitted Materials during Tests	Course book, a 2-page (4 sides) formula sheet and a calculator.	
Remarks	English is the language used in the course and in the exam.	
Elective	Yes	
Tags	Energy	
Targetgroup	Students from the Offshore Wind Energy Minor (being a core course from the minor) and all other students (selected as an elective course), to get an overview and understanding of how energy systems, i.e. the whole energy-based supply chain, are designed, and how these can and have to be transformed towards a sustainable future. The course provides basic knowledge for strategic and tactical planning. It is an overarching course that goes into energy consumption, electricity generation, sustainability, and climate change mitigation. Please, do not select this course if you have already attended the course TB142EA, as some content is similar.	
Category	BSc level	

Year

Organization

Education

2023/2024

Aerospace Engineering

Minors Aerospace Engineering

LR-MI-228-212 Airport Development	
Minor Coordinator	Ir. P.C. Roling

AE1110-I	Introduction to Aerospace Engineering I	5
Responsible Instructor	Prof.dr.ir. J.M. Hoekstra	
Instructor	Ir. J.A. Melkert	
Instructor	Prof.dr. F. Scarano	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	6/0/0/0 (instruction) + 2/0/0/0 studio class room	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This first part of the course Introduction to Aerospace Engineering presents an overall picture of the aeronautics domain. This overview involves a number of different perspectives on the aerospace domain, and shows some basic principles of the most important concepts for flight. Then the basic aerodynamics are covered, followed by flight mechanics.	
Study Goals	General knowledge of aircraft and their systems (such as avionics & propulsion) and its history Derive & apply atmospheric calculations Know and apply forces on an aircraft: Lift, drag, thrust. Derive power & efficiency equations. Derive & Apply equations for moments, stability & apply on aircraft configuration Structure & Materials: stress and dimensions also for non-unidirectional composites Apply control volume approaches Explain flow regimes (viscous/non-viscous; compressible/incompressible aerodynamics) and to estimate viscous and thermal effects Derive & apply equations of motion to determine aircraft performance in steady gliding, horizontal and climbing flight Derive aircraft performance diagram and flight envelope, in relation to aircraft morphology, lift-drag polar and engine performance	
Education Method	Lectures (live streamed, links on BrightSpace)	
Literature and Study Materials	Introduction to Flight, by John D. Anderson; 6th edition or later; ISBN of 7th edition: 9780071086059 Handouts	
Assessment	Exam consists of three different tests. Each of the tests are an on-site exam. The format of the tests is written (parts 1 & 2) and digital (part 3) exam in three separately scheduled parts. The three different parts of Intro to AE I will be graded separately, but each tests needs at least a 5.0 (so 4.75+ unrounded) to count for the exam. Each part can be redone separately. There is also a request to schedule an extra resit in January, but not decided when these data was entered.	
Set-up	Together with the course "Introduction to Aerospace Engineering II" (AE1110-II) this course is strongly related to the Project "Exploring Aerospace Engineering" (AE1111-I) which is running parallel in the first semester.	

AE3501-19	Air Transportation	3
Responsible Instructor	Ir. P.C. Roling	
Instructor	Dr.ir. B.F. Lopes Dos Santos	
Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course offers an introduction to the various aspects of air transportation, including air law, regulatory frameworks, airline economics, airline network and fleet planning, air transport safety, airport development, environmental impact, and air traffic management.	
Study Goals	Identify the characteristics and trends in the airline industry Explain the role of air transport in the global economy Identify and describe the economic and technical regulations and standards pertaining to organisations within the air transport industry. Identify and describe the structure of airline costs Describe the economic concepts of supply, demand, pricing and market structures Explain the key characteristics of different airline business models including the role of the airport. Explain the different network and fleet strategies adopted by airlines Explain key demand and supply factors affecting the air cargo sector	
Education Method	Lectures with possibly some exercises.	
Literature and Study Materials	R. Doganis, Flying off course, the economics of international airlines, London, Routledge, 2010, 4th Ed.	
Assessment	Digital exam	

AE3502-14	Airport Planning, Design and Operations	4
Responsible Instructor	Ir. P.C. Roling	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course Airport Planning, Design and Operation aims to provide students with an understanding of the complex interrelationships and interactions among airport capacity, airport demand, policy changes, investments, and environmental issues and the effects that changes in any of these can have on airport profits and performance	
Study Goals	Being able to Estimate the capacity of any airport configuration and understand the influence of weather, aircraft mix, and other operational parameters on capacity Determine runway requirements for a given aircraft type Estimate the (future) demand for transport of both passengers and freight Estimate the flight delays at an airport given the capacity of the facility and daily demand factors Analyze the noise impacts of aircraft in the vicinity of airports using computer models Estimate the geometric design characteristics of an airport including taxiways, aprons and runways Estimate siting criteria for new airports including terminals Conduct a Cost-Benefit analysis for new airport infrastructure developments Present and defend the policy recommendations consistent with the long-term vision for the airport	
Education Method	Lectures	
Reader	Richard de Neufville and Amedeo R. Odoni, Airport Systems: Planning, Design, and Management, McGraw-Hill, 2003	
Assessment	Computer-based exam (80% of grade) and homework assignments (20% of grade)	

AE3503	Strategic Planning for Airport Systems	6
Responsible Instructor	Ir. P.C. Roling	
Instructor	Dr. L. Li	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The airport business is high-paced, dynamic and unpredictable. Airport decision makers are faced with the continual challenge of having to make strategic plans and business decisions regarding infrastructure, market positioning, and commercial activities at the airport. The final project will explore the various aspects of strategic planning for airport systems	
Study Goals	The major goal of the capstone project is that it aims to provide students with an understanding of the complex interrelationships and interactions among airport capacity, airport demand, policy changes, investments, security and environmental issues and the effects that changes in any of these can have on airport profits and performance. Using a policy analysis/system-modeling framework, the full range of airport strategic planning issues will be addressed.	
Education Method	The final project involves a lab exercise in which each team of (5) students is asked to design and operate your own airport in a simulated environment based on the in-house developed Airport Business Suite (ABS). ABS is an integrated computational platform that will support policy and business decisions relating to airport development, through an integrated impact analysis. The lab exercise is directly linked to the lectures of the course Airport Planning, Design and Operation. Next to this, other tool s may also be part of the project.	
Course Relations	AE3502 must have been followed before taking part.	
Literature and Study Materials	Richard de Neufville and Amedeo R. Odoni, Airport Systems: Planning, Design, and Management, McGraw-Hill, 2003	
Assessment	The final project is to be concluded with a written report, along with a presentation in a mini-symposium. There is no written exam. The project is graded based on the quality of the report. A peer evaluation will be used within each group.	

CT3080LR	Landside accessibility of Airports	6
Responsible Instructor	Dr.ir. A.M. Salomons	
Responsible Instructor	Dr.ir. A. Gavriilidou	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course offers an introduction to all aspects of landside accessibility through assessing performances of transport modes and operational systems, related to transport to and from the airport. This includes elements of infrastructure and transport service networks, for which have to be considered: demand, capacity, spatial layout, economics, environmental and social impacts, local infrastructural development, and the complex integration of airports into regional and national transport system(s). Amongst others, the lectures will contain: Airport and airport systems and operations, including crowd monitoring and management in airports; Landside access modes and systems related to airports: the conventional and advanced road & rail-based components, its operations and performances; Performances of airport access modes, considering: infrastructural, technical/technological, operational, economic, environmental, social and policies; Control and management of landside access modes: introduction into ITS (Intelligent Transport Solutions); Planning and design of landside access modes and systems; Developing an airport as the multimodal transport node.	
Study Goals	After successful completion of the course, the student is able to: identify the characteristics and performances of transport modes providing landside access to an airport; analyse, model, evaluate, plan, and design of airport landside access modes; analyse and discuss the role of airports as multimodal transport modes in transport networks; analyse and discuss urban planning and developments around airports.	
Education Method	Lectures and assignment	
Assessment	1) Written exam in ANS (65%) 2) Assignment (to be handed in before the exam) (35%)	
Tags	Building & Spatial Development Modelling Policy Analysis Project planning / management Research Methods Transport & Logistics	
Expected prior Knowledge	No	
Academic Skills	Students are expected to exercise the following academic skills: Interpretation, writing review, judgemental skills, reasoning / arguing, writing concise and clear reports.	
Literature & Study Materials	BrightSpace postings of lecture notes and presentations; book "Landside Accessibility of Airports" by Milan Janic	
Judgement	1) Written ANS exam (65%) 2) Assignment (to be handed in before the exam) (35%) (5% - Midterm presentation; 5% - Group peer review; 25% - Assignment report)	
Permitted Materials during Exam	To be announced	
Collegerama	No	

IO3818	Designing an Airport Ecosystem	6
Course Coordinator	Ir. M.F. Beets	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The course Designing in Airport Ecosystem is built around an airport system and mobility. It is a research and design course with Schiphol as the main stakeholder, one of the busiest, main airports in Europe with around 70 million passengers a year. Other stakeholders join as well. You work on a real case for a client in mobility or airport-related business with different stakeholders and their interests in passengers. It is a human-centered design course. You study the airport and passengers from home to the destination and back. You will explore touchpoints in this journey. An important part is doing research with users, you will find hick-ups in experience, that can be improved. You may create a seamless passenger journey using current technology that meets the interests of the user and the client and relevant other stakeholders. Depending on the case you will design product/services like digital service, baggage solutions or solutions to diminish queuing, at the gate or catering points. You work in a group of 4 students of different faculties in a multi-disciplinary team.	
Study Goals	You will get insight into the challenges of what is showing up while designing operations at an airport. Learning about the different stakeholders involved and their needs/interests. You will get insight into designing from a passenger perspective. You will meet and learn about airport operations. Learning goals : LO1: analyze the context and the role of stakeholders within an airport ecosystem and create a framework for design. LO2: to integrate desirable, feasible, and viable aspects in a concept proposal and prototype. LO3: to apply the double diamond innovation process and tools and methods to design. LO4: to present relevant issues and design decisions of your work in an argued and concise way to stakeholders and others. LO5: to reflect on the process and evaluate the results in relation to the case owner and the design goal and criteria.	
Education Method	The course is structured into 3 parts. Part 1- Working on your project brief and exploring research and observing interactions, to find and define challenges and user needs/ problems in the context. You will end up with a design brief that includes a (re)defined problem, a context, a target group to design for, and first ideas. Part 2 is a developing phase, generating ideas in more detail and developing one into a concept. Part 3 -You develop a prototype and or movie of a service/product that meets the requirements of the client, the user, and other stakeholders. You will finally present your work to the client, peers, and coaches.	
Literature and Study Materials	Relevant books and reading materials will be presented throughout the course. Links will be available on Brightspace.	
Assessment	Your final individual grade will be assessed by using a rubric. You will deliver: -frame report (research findings and design brief) -design report (iteration and design results) -presentation and prototype -reflection	

TB241TB	Logistics 2	5
Module Manager	Dr. S. Fazi	
Co-responsible for assignments	I. Dursun	
Contact Hours / Week x/x/x/x	4/0/0/0 excl 10 hours of self study	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	According to the Council of Supply Chain Management Professionals logistics is defined as the process of planning, implementing, and controlling procedures for the efficient and effective transportation and storage of goods including services, and related information from the point of origin to the point of consumption for the purpose of conforming to customer requirements. Logistics activities should be in line with other business activities such as marketing and purchasing, in order to create sustainable competitive advantages for individual companies and service providers, and for the supply chain as a whole. Given the strategic importance of logistics for individual organizations and supply chains, it is vital to know how the performance of logistics activities is measured, and how these activities can be modelled and improved. Therefore in this course different logistics aspects are discussed and analyzed. A selection of quantitative methodologies are introduced to model and provide solutions to a wide range of relevant problems in logistics. We focus on two classes of logistics problems: Logistics problems within organizations: forecasting and demand modeling; inventory management; clustering and classification of items. Logistics problems between organizations: distribution and transportation problems. Methodologies include analytical models, linear programming models, basic data mining techniques.	
Study Goals	At the end of this course you will be able to: Analyze relevant aspects of logistics and transportation system: inventory, routing, location, data; Formulate (and solve) a selection of relevant logistics problems; Select and apply an appropriate methodology for a given logistics problem; Find possibilities to improve/adapt a logistics system through data and analytical models; Model operational aspects through mathematical linear programming models; Understand the integrative role of logistics function in and between organizations;	
Education Method	Lectures Assignments Self-study The educational activities will be held on campus, and attendance is highly recommended to complete the course. Consultation hours will be held both online and on campus.	
Literature and Study Materials	A selection of chapters from: 1) Reid, R. D., Sanders, N. R., Operations Management: An Integrated Approach, 5th edition, Wiley, 2013. 2) Daganzo, C.F., Logistics Systems Analysis, 4th edition, Springer-Verlag, 2010. 3) Zaki, Mohammed J., Wagner Meira Jr, and Wagner Meira. Data mining and analysis: fundamental concepts and algorithms. Cambridge University Press, 2014. 4) A booklet provided at the start of the course 5) Slides and exercises	
Prerequisites	Logistiek 1, Basic mathematics	
Assessment	The course consists of two parts with the following weighting: Logistics problems within organizations: 50%. Distribution problems: 50%. Grading: One written exam: 80%. This is a closed-book exam with the possibility of bringing one formula sheet created by the student on a single A4 paper. The A4 can be written on both sides. 2 Group Assignments (one for each part of the course): 20%. Expected group size: 4 students. The passing grade is 5.8, calculated as: $0.8 * (\text{final exam}) + 0.2 * ((\text{Assignment 1} + \text{Assignment 2})/2)$. Minimum grade for final exam: minimum grade is 5.0 Minimum grade for assignments: average grade at least 5 $\rightarrow (\text{Assignment 1} + \text{Assignment 2})/2 \geq 5.0$ It is possible to repair the assignments, but the maximum grade that can be obtained for each repaired assignment is 6. The repair	

can also be performed individually, by resubmitting every deliverable.

Any sufficient mark for any assessment can be retained for the next academic year.

In case of unforeseen circumstances, the final exam will take place in an online format: a take-home exam with a limited duration. The student will receive the exam via email at the start of the exam session and must upload the answers in Brightspace within the time limit. In case the regular on-campus exam can be held, no online possibility will be offered.

Exam Hours

3 hours

Year	2023/2024
Organization	Aerospace Engineering
Education	Minors Aerospace Engineering

LR-MI-232-23 Space Missions	
Minor Coordinator	Dr.ir. J. Bouwmeester
In association with the Faculty of	CITG LR
Administration by the Faculty of	LR

Year	2023/2024
Organization	Aerospace Engineering
Education	Minors Aerospace Engineering

Standard programme	
Minor Coordinator	Dr.ir. J. Bouwmeester

AE3522UL	Fundamentals of Astronomy	4
Responsible Instructor	M.A. Kenworthy	
Instructor	Dr. M. Rybak	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course provides 2 lectures/instructions per week of 2x45 minutes each. There are three subtopics consisting of Stars, Galaxies and Exoplanets. Each section lasts 2 weeks (with 4 lectures).	
Study Goals	The aim of this course is to give an overview of astronomical and astrophysical phenomena appropriate for the Space Missions Minor. The learning goals are: Understand the origin, evolution and end stages of stars Understand the structure and evolution of galaxies The evolution of the Universe and galaxies over cosmological timescales How we detect extrasolar planets and the search for life in the Universe	
Education Method	The content is provided in lectures throughout the course and with homework sheets every two weeks.	
Literature and Study Materials	Lecture notes available on BrightSpace. Astronomy papers that can be accessed online by the students.	
Prerequisites	This course is only available for students who are registered in the Space Missions minor.	
Assessment	Three assignments (one on each of the topics of Galaxies, Stars and Exoplanets) will be given at the end of that subsection of lectures with a deadline one week after the end of the section. Each will count for 15% of the final grade, for a total of 45% of the final grade. If you fail to submit the homework on time, you get a second opportunity (a resit) to hand the homework in up to one week later, and it will receive only 75% of the score of that specific homework. After one week, the homeworks will be marked with a score of 1. It is not possible to retake a homework based on a low grade. A student needs to compensate with the other homeworks and the exam. The final exam counts for 55% of the final grade. If you fail the course (homeworks + final exam), you can take a retake exam. The best grade of the two exams will count as the final grade for the exam.	

AE3523	Planetary Sciences	3
Responsible Instructor	Dr. S.J. de Vet	
Instructor	Dr. S.M. Cazaux	
Instructor	M.A. Kenworthy	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course Planetary Science will offer you a perspective on the state-of-art in the field of planetary science performed inside the solar system. Past missions and insights from ongoing missions provide a unique perspective on the research performed in this field and new arising science questions. In this course we will reflect on the cross-disciplinary character of the science being performed at the interface of geoscience and astronomy.	
Study Goals	After completion of the course Planetary Science, you will be able to explain how we explore the Solar System and you will apply your scientific knowledge to describe characteristics of planets inside the Solar System and the exoplanets beyond. After this course the student is (cq. you are) able to: LO1: Explain how knowledge on planetary processes and evolution is obtained using planetary exploration missions. LO2: Perform calculations on planetary processes that have been observed inside the Solar System. LO3: Analyse key properties of planets and other planetary bodies inside the Solar System, using data from past and present satellite missions. LO4: Point out the underlying reasons for the physical differences between planets and other planetary bodies inside the Solar System.	
Education Method	Lectures (possibly pre-recorded with online/on-site option for Q&A after streaming of the lecture); two learning labs (i.e., homework assignments), with one dedicated to planetary processes and materials, and one focussed on icy moons, exomoons and habitability.	
Literature and Study Materials	The course and lectures will be based on parts of the highly recommended book: McSween, Jr, H., Moersch, J., Burr, D., Dunne, W., Emery, J., Kah, L., & McCanta, M. (2019). Planetary Geoscience. Cambridge: Cambridge University Press. doi:10.1017/9781316535769 Lecture notes (slides) and assignments will be made available on BrightSpace.	
Assessment	The two learning labs are offered as a homework assignment via BrightSpace and will be graded. Feedback to the student will be provided after grading. A retake for the assignment is possible until 3 weeks after the first submission deadline in the following two cases: (1.) when you miss the deadline (0.2 points will be subtracted per day from the final grade) and (2.) when you fail the assignment you can use the feedback to improve and resubmit the assignment to a maximum grade of 6.0. For all students the final grade for the course will be based on a written exam (60%) and the grades for the learning labs (20% each). The exam will test all theory provided in the lectures slides, the literature (via self-study) and the skills practiced in the provided exercises.	
Enrolment / Application	This course is primarily intended for students who are registered in the Space Missions minor. If you would like to enrol for this course as elective outside this scope (e.g. for a 'free minor'), please send an email to course coordinator dr. Sebastiaan de Vet (s.j.deviet@tudelft.nl) and motivate your request. The course coordinator will decide if you can be admitted.	

AE3524	Spacecraft Design and Astrodynamics	5
Responsible Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr.ir. P.N.A.M. Visser	
Instructor	Dr. J. Guo	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course comprises 20 lectures. Some lectures have more emphasis on theory, some more on exercises. Below is a list of the lectures, the lecturers and its content. Please note that the order of the lectures is indicative and is subject to change. Up-to-date information will be provided on BrightSpace. 1. Spacecraft Basics - Jasper Bouwmeester Brief history of spaceflight Principles and examples of spacecraft Subsystem breakdown & functions 2. Astrodynamics 1/2 - Pieter Visser Brief history orbital mechanics Keplers law Newtons universal law of gravitation Restricted two-Body problem Orbital elements 3. Astrodynamics 2/2 - Pieter Visser Application orbits Ground tracks Orbit perturbations Orbit transfer 4. Structures 1/2 - Ines Uriol Balbin Spacecraft design loads Material selection for space missions Simulation of space structures Testing of space structures 5. Structures 2/2 - Ines Uriol Balbin Structural design Structural analysis 6. Command & Data Handling 1/2 - Alessandra Menicucci Intro to On-Board Data Handling (OBDH) Scope of an on-board computer Data handling system & unit architecture OBDH bus architecture OBDH operations modes 7. Command & Data Handling 2/2 - Alessandra Menicucci Designing an On-Board Computer Radiation effect mitigation Data protection On-Board Computer processor selection Memory selection 8. Electrical Power 1/2 - Jasper Bouwmeester Energy sources Solar power acquisition Voltage regulation Power distribution 9. Electrical Power 2/2 - Jasper Bouwmeester Preliminary sizing Design 10. Thermal Control - Jasper Bouwmeester Thermodynamic basics Passive control Active cooling & heating Sizing and design 11. Attitude Determination & Control 1/2 - Jasper Bouwmeester Orientation & reference frames Sensors overview & working principles Actuators overview & working principles Algorithms: general principles 12. Attitude Determination & Control 2/2 - Jasper Bouwmeester Preliminary sizing Design of sensors Design of actuators	

13. Propulsion 1/2 - Angelo Cervone
Rocket performance equations
Liquid and solid propellant engines
14. Propulsion 2/2 - Angelo Cervone
Electric propulsion
Advanced propulsion systems
15. Communication 1/2 - Stefano Speretta
Basic link budget
Main radio components
Communication chain
16. Communication 2/2 - Stefano Speretta
Link budget error and statistical analysis
Introduction to radio-navigation
17. Guidance, Navigation & Control 1/2 - Jian Guo
Guidance and trajectory planning
Navigation
Kinematics and dynamics
Trajectory control
18. Guidance, Navigation & Control 2/2 - Jian Guo
GNC system architecture design
19. Satellite Design - Guest lecturer
Example(s) of satellite design

Study Goals

The aim of this course is to provide insight in the functions and design options of spacecraft and its astrodynamics. The learning goals are:

Describe spacecraft orbits and calculate its main characteristics.
Formulate the technical principles of spacecraft bus subsystems.
Identify and compare the available subsystem options at component level, calculate their key characteristics and provide their physical and technical limitations.
Characterize the key performance parameters of different spacecraft subsystems and compare the values obtained by ideal theory to the real-life ones.

Education Method

The content is provided in lectures (three per week), of which some contain exercises to be performed, guided and explained in class. Students can self-study by reading annotated hand-outs and performing online exercises. Students will perform one mandatory test for each of the nine topics. Furthermore, AE and astronomy students will perform individual assignment(s) in which they will perform preliminary design of a satellite subsystem or a satellite orbit. This assignment is provided and guided by the associated lectures.

Literature and Study Materials

Annotated lecture slides, available on BrightSpace.
In class exercises. These are part of the lecture slides.
Exercises in ANS for self-study.

Assessment

Recommended book: Wertz and Larson, Space Mission Analysis & Design.

The assessment comprises a (1) digital exam and (2) topic tests for all students. Only for AE and Astronomy students, this is complemented with (3) assignments.

1. Digital Exam

For all students, except AE students, the final grade of AE3524 will be for 100% based on a digital exam in ANS, to be taken in an allocated digital examination room on campus. The exam will test all theory provided in the lectures and skills practiced in the provided exercises and tests. A resit opportunity is provided at the end of the 2nd quarter.

2. Topic Tests

The following setup and rules apply:

Each student is required to complete tests for each of the 9 main topics of the course.
Students who have not completed all tests before the start of the exam will be excluded from the exam.
The tests are taken in ANS and can be completed in multiple sessions.
Each test is estimated to take between 45 and 90 minutes to complete. This is indicative; there is no time limit applied to the duration of the test.
Each test will open at the first lecture of the given topic. The lecture will indicate which questions can already be completed after the first lecture.
A bonus of +0.1 will be applied to the final grade per test which is completed within one week after the concluding (typically second) lecture of each topic with a result of 6.0 or higher.
Another bonus of +0.1 will be applied to the final grade if all 9 tests are completed in time with sufficient result, which would yield a total bonus of +1.0 in that case.
No exemptions or deviations from the rules above will be made for unforeseen circumstances, such as illness or other force majeure.
The questions in the tests are representative for the type and level of questions in the exam. They can be retaken for rehearsal.

3. Assignments (only AE & Astronomy students)

AE students: to account for the partial overlap with the BSC, the examination will count for 60% and one assignment will be provided which counts for 40% of the final grade.
Astronomy students: as alternative to the astronomy course, two assignments will be provided and included in the 4 ECTS extension of this course (AE3524E). Each assignment counts for 50% of its final grade.

The student performs preliminary design of a subsystem or algorithm, including calculations/simulations, according to a set of provided technical requirements. Assignments will be based on one of the nine topics and students can indicate their preferences. Assessment will be based on a small report and a 30-minute discussion with the responsible lecturer on the results. All assignment will be provided in the first lecture week and needs to be completed latest in the final (7th) lecture week. The following rules apply:

The penalty for exceeding the submission deadline, without prior permission from the supervisor or course coordinator, is a subtraction of 0.1 on the total assignment grade per working day beyond the deadline.
In case of a force majeure, the supervisor needs to be contacted as soon as possible. A shift of deadline will be provided if this is required and justified.
When the grade of the assignment is 6.0 or higher, you will not be given the opportunity to improve or do another assignment.
When the grade is below 5.0, the student must improve the work to be able to pass the course. Between 5.0 and 5.9, the student can choose.
For potential rework, the supervisor will indicate what needs to be improved. The student then has two more weeks to submit

Enrolment / Application	the improvement.
	The assignment grade is the average of the original and the improved work if this leads to a grade of > 6.0, or it becomes the highest grade if it is 6.0 or lower.
	Only one improvement opportunity will be given. If this is still below 5.0, a new assignment needs to be performed in the next academic year.
	This course is only available for students who are registered in the: (1) Space Missions minor (2) AE Space Engineering MSc track who have not followed an Aerospace Engineering BSc nor the Space Missions minor before and want to use this course as elective to get a quick understanding of space engineering.
It is not possible to follow this course through other programs (free minor, Erasmus program, master, etcetera).	

AE3529	Space Instrumentation	5
Responsible Instructor	Dr. J.J.D. Loicq	
Instructor	J.G. De Teixeira da Encarnacao	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	M.A. Kenworthy	
Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Space instruments are the links between the science target, the missions, the data-users, the spacecraft. The goal of this course will be to give students insights into the capabilities of instrumentation and how they contribute to the observational strategies of the mission. Space instrumentation conception is based on very multidisciplinary technics and technologies. Optics routes photons to detectors, when mechanics gives the stability and holds optical surfaces, detectors, radiators, electronics, while thermic is given by environment (orbits, spacecraft,) and apply constraints. The conception of the instruments in a system engineering approach will be discussed. Space Instrumentation can be subdivided into two main groups: remote and in situ instrumentations. Both will be covered during this course. The space instrumentation course will also give an overview of the already existing and under development instruments. The relationship between instruments and their associated space mission will be taught.	
Study Goals	The students should understand and describes The links between instruments, mission and observation strategies The physical observables (light, ions, magnetic/electric fields and related phenomena) the space instruments are measuring. The different type of space instruments. How they are designed and built The effect of the space environment on the instrument designs The keys drivers of a space instruments	
Education Method	The content is provided in lectures, which include some exercises that will be performed, guided , and explained. Students can self-study by reading lecture notes and performing online exercises on BrightSpace. Students will perform an individual assignment in in which they analyse an existing space mission.	
Assessment	For all students, the final grade will be for 80% based on a written exam and 20% based on an individual design assignment. Exam (80%): The exam will test all theories provided in the lectures and skills practiced in the provided exercises. Assignment (20%): In the assignment, it will be asked to the students to provide a critical study of a space mission by the mean an analysis of (at least) one scientific article related to a mission (e.g. Space Science Review, JATIS,) The objective will be to understand the space instrument chosen and its relationship with the mission, and their observation strategies. Assessment will be based on a small report (max 6 pages).	

AE3538	Space Mission: Engineering Case Study	5
Responsible Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Space missions are extremely complex projects. They require the development of both the spacecraft and the payload in a coordinated manner and, at the same time, fulfilling many boundary conditions like budget, time scales, orbit, weight of the mission and others. This task is usually done by a large team of professionals with an ample variety of backgrounds, skills, and expertise. In order to create a certain flavour of these collaborations, the Mission Assignment in the LDE Space Minor was designed. There, each student team will work on a specific Mission Assignment, provided by a Mission Client (e.g. ESA, SRON, TNO, etc.). This Mission Client will provide the mission objectives and top-level requirements. Certain top level requirements from the Mission Client will be used to design a spacecraft, able to fill full the scientific goal of the mission. The scientific mission will be fulfilled by the design of a payload in the course AE3539UL. This course is linked to the design of the spacecraft to house this payload. Key options for each subsystem and component will be identified, either commercially available or custom designed. A trade-off for each will be performed and justified. All system budgets will close and interfaces are compatible, as proven in tables and figures.	
Study Goals	During this course, each team provides a preliminary design of the spacecraft or space system. The goal of the course is to make the students more familiar with the design process of a spacecraft. More specific, the spacecraft requirements to be set shall be driven by the functional analysis of the spacecraft and the mission profile using all kind of inputs (e.g. literature study, practical examples, etc.). Based on this, the initial spacecraft design can be done (sizing, budgeting, etc.), taking the requested payload into account (course AE3539UL). As a last step, the different spacecraft subsystems are designed (analysis and architecture).	
Education Method	Each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. Students will partially use the Collaborative Design Lab (CDL) (first session physical presence mandatory) in order to experience the use of such a facility. The course will start with a kick-off meeting. Physical presence during this meeting is mandatory. During the course, the students will meet with their team mates and tutors regularly. Physical presence is expected during those sessions. You may skip maximum two sessions due to absence. If you are absent during more than those two sessions, you cannot pass this course during this academic year.	
Assessment	Throughout the course, several review processes will take place (physical presence mandatory). The other teams, the Mission Client and members of the Academic Staff will participate in these reviews. More specifically, the global assessment will be the combination of the outcome of oral presentations (physical presence mandatory), and group and individual reports. If the result is considered insufficient (<6.0), a re-attempt will be offered which consists of a rework of the individual and group report, within one week after the final assessment moment. Additionally, the student needs to do a presentation with a summary of the progress on the mission spacecraft design. The finalization of both parts (report rework and presentation) take place one week after the final assessment moment.	
Enrolment / Application	For Space Missions minor students only	

AE3539UL	Space Mission: Scientific Case Study	5
Responsible Instructor	Dr. A. Monreal Ibero	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Space missions are extremely complex projects. They require the development of both the spacecraft and the payload in a coordinated manner and, at the same time, fulfilling many boundary conditions (e.g. budget, time scales, orbit, weight of the mission and others), that often implies the need to look for innovative solutions out-of-the-box that do not appear in textbooks. Moreover, this is usually done by a large team of professionals with an ample variety of backgrounds, skills, and expertise, that need to understand each other and work together for the common objective, even if they do not always speak the same (technical and scientific) language. Future professionals in the aerospace industry need to learn how to work under these conditions, yet this is quite far from what a (science or engineer) student can find in a typical bachelor course. The Mission Assignment in the Space Minor was designed to fill this gap. There, each student team (consisting of about 8 students) will work individually on a Mission Assignment, provided by a Mission Client. The latter can be ESA, SRON, TNO, etc. This Mission Client will provide the mission objectives and top-level requirements. The Space Mission: Scientific Case Study course (hereafter SCS) covers the scientific part of the mission: the payload. Here, the students team translates this into top-level payload design, first. Then, as the course progresses, the parameters of the required observations are determined and fine-tuned. Specifically, instrumental parameters such as, for instance, the central observing wavelength, the total bandwidth of the observations, the spectral resolution, spatial sampling, field of view, or required temporal sampling are considered. These parameters then determine the choice of telescope aperture (to gather enough photons) and detector technology used. In short, the students define the instrumental configuration required to meet the science goals. The course is complemented and should be done together with the twin course Space Mission: Engineering Case Study (code: AE3538), that deals with the spacecraft.	
Study Goals	During the Project Design and Analysis course (see AE3526), each student team worked out an approach on the first scientific analysis and conceptual system design. Based on this, each team provides a preliminary scientific case and key question to be answered, as well as top level requirements for the instrumentation in the spacecraft or space system. The goal of the course is to make the students more familiar with the processes involved in the definition of the scientific aspects of a spatial mission. In particular, the students will find and review the relevant literature on a given science topic. They will need to get a general picture on the science topic and a more detailed insight of the specific question(s) they will address with the mission. Likewise, they will identify similar state-of-the-art missions (or ground-based facilities), and determine and highlight the unique features of this mission. Then, they will establish and investigate the physical relationships between the science target, the optomechanics within the spacecraft, and the resultant detection and processing of the detected signal. In the process, the low level scientific and technical requirements from the initial mission statement and requirements, are defined and refined in an iterative manner.	
Education Method	Each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. The mission, as whole, will start with a kick-off meeting (mandatory attendance). During the course, the students will meet with their team mates and the tutors regularly. Regarding the SCS, the students will have the The Studio Classroom in the Huygens building (Leiden University premises), available one whole day per week for that purpose.	
Assessment	Throughout the course, several review processes will take place. The other teams, the Mission Client and members of the Academic Staff participate in these reviews. More specifically, the global assessment will be the combination of the outcome of oral presentations (mandatory presence), and group and individual reports. Should the student fail to pass any of the items needed for the evaluation, there will be an opportunity for resit in the week after the deadline for submitting the reports (written part), or the week after presentation days (oral part). Details on the format and content for the both the main and resit evaluation procedures will be found on Brightspace.	

CT3533	Earth Observation	3
Responsible Instructor	Dr.ir. J. Timmermans	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Earth observation is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Remote sensing observations from spaceborne instrument inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. This course aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain bio- & geophysical variables iii) to evaluate the different advantages/disadvantages of earth observation technologies. This should allow students to understand how to formulate user requirements for understanding/monitoring Earth into an earth observation mission requirements.	
Study Goals	The overall goal of this course is to provide insight into the physical principles of Earth observation and how earth observation technologies can be used to assess and monitor geo- & biophysical variables on Earth. After this course the student is (cq. you are) able to: LO1: Summarise the signals of the earth in the different parts of the electromagnetic (EM) spectrum and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and subsurface). LO2: Characterise the different observable signals and parameters from Earth and how they can be observed using different remote sensing technologies. LO3: Assess remote sensing technologys advantages/disadvantages for observing geo- & biophysical variables on Earth. LO4: Translate user requirements into mission requirements for earth observation mission. LO5: Evaluate a basic remote sensing analysis to retrieve a geo- & biophysical variable from Earth Observation data.	
Education Method	Classical lectures complemented with pre-recorded videos to maximize interaction in class (flip-the-classroom). Two lab sessions to work on assignment in groups with peer-review.	
Assessment	The students are assessed on their comprehension of the educational material through an assignment. Students are required to perform an assignment where they translate user requirements into mission requirements for a specific earth observation application. Specifically, the students will choose a societal or geophysical topic of interest and area of interest (AOI) to formulate a research question. Select an appropriate satellite mission and get access to the data of this mission Test and select the data analysis platform Analyse the data (this may require some coding) Report your findings in a final report and presentation Through review of the end-of-assignment presentation and final report a grade is given. Because of the nature of the assignment, the assignment can only be performed during the execution of the program.	
Enrolment / Application	This course is intended for students who are registered in the Space Missions minor. If you would like to enroll for this course as elective outside this scopce (e.g. for a 'free minor'), please send an email to course coordinator Joris Timmermans and motivate your request. The course coordinator will decide if you can be admitted.	
Expected prior Knowledge	No specific knowledge required.	
Academic Skills	-	
Literature & Study Materials	Video recordings Book: Rees (2012). Physical Principles of Remote Sensing https://doi.org/10.1017/CBO9781139017411	
Judgement	Students need to score minimal a 6 for the assignment to pass the course. If students do not obtain a passible grade for the assignment the lecturer decides how the deliverable can be amended or remade. For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2023/2024
Organization	Aerospace Engineering
Education	Minors Aerospace Engineering

Programme for Astronomy students (Leiden University)	
Minor Coordinator	Dr.ir. J. Bouwmeester

AE3523	Planetary Sciences	3
Responsible Instructor	Dr. S.J. de Vet	
Instructor	Dr. S.M. Cazaux	
Instructor	M.A. Kenworthy	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course Planetary Science will offer you a perspective on the state-of-art in the field of planetary science performed inside the solar system. Past missions and insights from ongoing missions provide a unique perspective on the research performed in this field and new arising science questions. In this course we will reflect on the cross-disciplinary character of the science being performed at the interface of geoscience and astronomy.	
Study Goals	After completion of the course Planetary Science, you will be able to explain how we explore the Solar System and you will apply your scientific knowledge to describe characteristics of planets inside the Solar System and the exoplanets beyond. After this course the student is (cq. you are) able to: LO1: Explain how knowledge on planetary processes and evolution is obtained using planetary exploration missions. LO2: Perform calculations on planetary processes that have been observed inside the Solar System. LO3: Analyse key properties of planets and other planetary bodies inside the Solar System, using data from past and present satellite missions. LO4: Point out the underlying reasons for the physical differences between planets and other planetary bodies inside the Solar System.	
Education Method	Lectures (possibly pre-recorded with online/on-site option for Q&A after streaming of the lecture); two learning labs (i.e., homework assignments), with one dedicated to planetary processes and materials, and one focussed on icy moons, exomoons and habitability.	
Literature and Study Materials	The course and lectures will be based on parts of the highly recommended book: McSween, Jr, H., Moersch, J., Burr, D., Dunne, W., Emery, J., Kah, L., & McCanta, M. (2019). Planetary Geoscience. Cambridge: Cambridge University Press. doi:10.1017/9781316535769 Lecture notes (slides) and assignments will be made available on BrightSpace.	
Assessment	The two learning labs are offered as a homework assignment via BrightSpace and will be graded. Feedback to the student will be provided after grading. A retake for the assignment is possible until 3 weeks after the first submission deadline in the following two cases: (1.) when you miss the deadline (0.2 points will be subtracted per day from the final grade) and (2.) when you fail the assignment you can use the feedback to improve and resubmit the assignment to a maximum grade of 6.0. For all students the final grade for the course will be based on a written exam (60%) and the grades for the learning labs (20% each). The exam will test all theory provided in the lectures slides, the literature (via self-study) and the skills practiced in the provided exercises.	
Enrolment / Application	This course is primarily intended for students who are registered in the Space Missions minor. If you would like to enrol for this course as elective outside this scope (e.g. for a 'free minor'), please send an email to course coordinator dr. Sebastiaan de Vet (s.j.deviet@tudelft.nl) and motivate your request. The course coordinator will decide if you can be admitted.	

AE3524	Spacecraft Design and Astrodynamics	5
Responsible Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr.ir. P.N.A.M. Visser	
Instructor	Dr. J. Guo	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course comprises 20 lectures. Some lectures have more emphasis on theory, some more on exercises. Below is a list of the lectures, the lecturers and its content. Please note that the order of the lectures is indicative and is subject to change. Up-to-date information will be provided on BrightSpace. 1. Spacecraft Basics - Jasper Bouwmeester Brief history of spaceflight Principles and examples of spacecraft Subsystem breakdown & functions 2. Astrodynamics 1/2 - Pieter Visser Brief history orbital mechanics Keplers law Newtons universal law of gravitation Restricted two-Body problem Orbital elements 3. Astrodynamics 2/2 - Pieter Visser Application orbits Ground tracks Orbit perturbations Orbit transfer 4. Structures 1/2 - Ines Uriol Balbin Spacecraft design loads Material selection for space missions Simulation of space structures Testing of space structures 5. Structures 2/2 - Ines Uriol Balbin Structural design Structural analysis 6. Command & Data Handling 1/2 - Alessandra Menicucci Intro to On-Board Data Handling (OBDH) Scope of an on-board computer Data handling system & unit architecture OBDH bus architecture OBDH operations modes 7. Command & Data Handling 2/2 - Alessandra Menicucci Designing an On-Board Computer Radiation effect mitigation Data protection On-Board Computer processor selection Memory selection 8. Electrical Power 1/2 - Jasper Bouwmeester Energy sources Solar power acquisition Voltage regulation Power distribution 9. Electrical Power 2/2 - Jasper Bouwmeester Preliminary sizing Design 10. Thermal Control - Jasper Bouwmeester Thermodynamic basics Passive control Active cooling & heating Sizing and design 11. Attitude Determination & Control 1/2 - Jasper Bouwmeester Orientation & reference frames Sensors overview & working principles Actuators overview & working principles Algorithms: general principles 12. Attitude Determination & Control 2/2 - Jasper Bouwmeester Preliminary sizing Design of sensors Design of actuators	

13. Propulsion 1/2 - Angelo Cervone
Rocket performance equations
Liquid and solid propellant engines
14. Propulsion 2/2 - Angelo Cervone
Electric propulsion
Advanced propulsion systems
15. Communication 1/2 - Stefano Speretta
Basic link budget
Main radio components
Communication chain
16. Communication 2/2 - Stefano Speretta
Link budget error and statistical analysis
Introduction to radio-navigation
17. Guidance, Navigation & Control 1/2 - Jian Guo
Guidance and trajectory planning
Navigation
Kinematics and dynamics
Trajectory control
18. Guidance, Navigation & Control 2/2 - Jian Guo
GNC system architecture design
19. Satellite Design - Guest lecturer
Example(s) of satellite design

Study Goals

The aim of this course is to provide insight in the functions and design options of spacecraft and its astrodynamics. The learning goals are:

Describe spacecraft orbits and calculate its main characteristics.
Formulate the technical principles of spacecraft bus subsystems.
Identify and compare the available subsystem options at component level, calculate their key characteristics and provide their physical and technical limitations.
Characterize the key performance parameters of different spacecraft subsystems and compare the values obtained by ideal theory to the real-life ones.

Education Method

The content is provided in lectures (three per week), of which some contain exercises to be performed, guided and explained in class. Students can self-study by reading annotated hand-outs and performing online exercises. Students will perform one mandatory test for each of the nine topics. Furthermore, AE and astronomy students will perform individual assignment(s) in which they will perform preliminary design of a satellite subsystem or a satellite orbit. This assignment is provided and guided by the associated lectures.

Literature and Study Materials

Annotated lecture slides, available on BrightSpace.
In class exercises. These are part of the lecture slides.
Exercises in ANS for self-study.

Assessment

Recommended book: Wertz and Larson, Space Mission Analysis & Design.

The assessment comprises a (1) digital exam and (2) topic tests for all students. Only for AE and Astronomy students, this is complemented with (3) assignments.

1. Digital Exam

For all students, except AE students, the final grade of AE3524 will be for 100% based on a digital exam in ANS, to be taken in an allocated digital examination room on campus. The exam will test all theory provided in the lectures and skills practiced in the provided exercises and tests. A resit opportunity is provided at the end of the 2nd quarter.

2. Topic Tests

The following setup and rules apply:

Each student is required to complete tests for each of the 9 main topics of the course.
Students who have not completed all tests before the start of the exam will be excluded from the exam.
The tests are taken in ANS and can be completed in multiple sessions.
Each test is estimated to take between 45 and 90 minutes to complete. This is indicative; there is no time limit applied to the duration of the test.
Each test will open at the first lecture of the given topic. The lecture will indicate which questions can already be completed after the first lecture.
A bonus of +0.1 will be applied to the final grade per test which is completed within one week after the concluding (typically second) lecture of each topic with a result of 6.0 or higher.
Another bonus of +0.1 will be applied to the final grade if all 9 tests are completed in time with sufficient result, which would yield a total bonus of +1.0 in that case.
No exemptions or deviations from the rules above will be made for unforeseen circumstances, such as illness or other force majeure.
The questions in the tests are representative for the type and level of questions in the exam. They can be retaken for rehearsal.

3. Assignments (only AE & Astronomy students)

AE students: to account for the partial overlap with the BSC, the examination will count for 60% and one assignment will be provided which counts for 40% of the final grade.
Astronomy students: as alternative to the astronomy course, two assignments will be provided and included in the 4 ECTS extension of this course (AE3524E). Each assignment counts for 50% of its final grade.

The student performs preliminary design of a subsystem or algorithm, including calculations/simulations, according to a set of provided technical requirements. Assignments will be based on one of the nine topics and students can indicate their preferences. Assessment will be based on a small report and a 30-minute discussion with the responsible lecturer on the results. All assignment will be provided in the first lecture week and needs to be completed latest in the final (7th) lecture week. The following rules apply:

The penalty for exceeding the submission deadline, without prior permission from the supervisor or course coordinator, is a subtraction of 0.1 on the total assignment grade per working day beyond the deadline.
In case of a force majeure, the supervisor needs to be contacted as soon as possible. A shift of deadline will be provided if this is required and justified.
When the grade of the assignment is 6.0 or higher, you will not be given the opportunity to improve or do another assignment.
When the grade is below 5.0, the student must improve the work to be able to pass the course. Between 5.0 and 5.9, the student can choose.
For potential rework, the supervisor will indicate what needs to be improved. The student then has two more weeks to submit

Enrolment / Application	the improvement.
	The assignment grade is the average of the original and the improved work if this leads to a grade of > 6.0, or it becomes the highest grade if it is 6.0 or lower.
	Only one improvement opportunity will be given. If this is still below 5.0, a new assignment needs to be performed in the next academic year.
	This course is only available for students who are registered in the: (1) Space Missions minor (2) AE Space Engineering MSc track who have not followed an Aerospace Engineering BSc nor the Space Missions minor before and want to use this course as elective to get a quick understanding of space engineering.
It is not possible to follow this course through other programs (free minor, Erasmus program, master, etcetera).	

AE3524E	Spacecraft Design and Astrodynamics Extended	4
Responsible Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr. J. Guo	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE3524	
Course Contents	This is the extension module for AE3524 Spacecraft Design & Astrodynamics.	
Study Goals	Perform preliminary design of a spacecraft subsystem and/or its astrodynamics based on key requirements and evaluate its performance	
Education Method	Only excercises. See AE3524 for more information.	
Assessment	Two extended assignments with a studyoad of 56 hours each which count for 50% of the final grade. Students can provide their preferences for the assignments.	
Enrolment / Application	This course is only available for students who are registered in the Space Missions minor and follow an Astronomy bachelor at the University of Leiden. It is not available for other students.	

AE3529	Space Instrumentation	5
Responsible Instructor	Dr. J.J.D. Loicq	
Instructor	J.G. De Teixeira da Encarnacao	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	M.A. Kenworthy	
Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Space instruments are the links between the science target, the missions, the data-users, the spacecraft, The goal of this course will be to give students insights into the capabilities of instrumentation and how they contribute to the observational strategies of the mission. Space instrumentation conception is based on very multidisciplinary technics and technologies. Optics routes photons to detectors, when mechanics gives the stability and holds optical surfaces, detectors, radiators, electronics, while thermic is given by environment (orbits, spacecraft,) and apply constraints. The conception of the instruments in a system engineering approach will be discussed. Space Instrumentation can be subdivided into two main groups: remote and in situ instrumentations. Both will be covered during this course. The space instrumentation course will also give an overview of the already existing and under development instruments. The relationship between instruments and their associated space mission will be taught.	
Study Goals	The students should understand and describes - The links between instruments, mission and observation strategies - The physical observables (light, ions, magnetic/electric fields and related phenomena) the space instruments are measuring. - The different type of space instruments. How they are designed and built - The effect of the space environment on the instrument designs - The keys drivers of a space instruments	
Education Method	The content is provided in lectures, which include some exercises that will be performed, guided , and explained. Students can self-study by reading lecture notes and performing online exercises on BrightSpace. Students will perform an individual assignment in in which they analyse an existing space mission.	
Assessment	For all students, the final grade will be for 80% based on a written exam and 20% based on an individual design assignment. Exam (80%): The exam will test all theories provided in the lectures and skills practiced in the provided exercises. Assignment (20%): In the assignment, it will be asked to the students to provide a critical study of a space mission by the mean an analysis of (at least) one scientific article related to a mission (e.g. Space Science Review, JATIS.) The objective will be to understand the space instrument chosen and its relationship with the mission, and their observation strategies. Assessment will be based on a small report (max 6 pages).	

AE3538	Space Mission: Engineering Case Study	5
Responsible Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Space missions are extremely complex projects. They require the development of both the spacecraft and the payload in a coordinated manner and, at the same time, fulfilling many boundary conditions like budget, time scales, orbit, weight of the mission and others. This task is usually done by a large team of professionals with an ample variety of backgrounds, skills, and expertise. In order to create a certain flavour of these collaborations, the Mission Assignment in the LDE Space Minor was designed. There, each student team will work on a specific Mission Assignment, provided by a Mission Client (e.g. ESA, SRON, TNO, etc.). This Mission Client will provide the mission objectives and top-level requirements. Certain top level requirements from the Mission Client will be used to design a spacecraft, able to fill full the scientific goal of the mission. The scientific mission will be fulfilled by the design of a payload in the course AE3539UL. This course is linked to the design of the spacecraft to house this payload. Key options for each subsystem and component will be identified, either commercially available or custom designed. A trade-off for each will be performed and justified. All system budgets will close and interfaces are compatible, as proven in tables and figures.	
Study Goals	During this course, each team provides a preliminary design of the spacecraft or space system. The goal of the course is to make the students more familiar with the design process of a spacecraft. More specific, the spacecraft requirements to be set shall be driven by the functional analysis of the spacecraft and the mission profile using all kind of inputs (e.g. literature study, practical examples, etc.). Based on this, the initial spacecraft design can be done (sizing, budgeting, etc.), taking the requested payload into account (course AE3539UL). As a last step, the different spacecraft subsystems are designed (analysis and architecture).	
Education Method	Each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. Students will partially use the Collaborative Design Lab (CDL) (first session physical presence mandatory) in order to experience the use of such a facility. The course will start with a kick-off meeting. Physical presence during this meeting is mandatory. During the course, the students will meet with their team mates and tutors regularly. Physical presence is expected during those sessions. You may skip maximum two sessions due to absence. If you are absent during more than those two sessions, you cannot pass this course during this academic year.	
Assessment	Throughout the course, several review processes will take place (physical presence mandatory). The other teams, the Mission Client and members of the Academic Staff will participate in these reviews. More specifically, the global assessment will be the combination of the outcome of oral presentations (physical presence mandatory), and group and individual reports. If the result is considered insufficient (<6.0), a re-attempt will be offered which consists of a rework of the individual and group report, within one week after the final assessment moment. Additionally, the student needs to do a presentation with a summary of the progress on the mission spacecraft design. The finalization of both parts (report rework and presentation) take place one week after the final assessment moment.	
Enrolment / Application	For Space Missions minor students only	

AE3539UL	Space Mission: Scientific Case Study	5
Responsible Instructor	Dr. A. Monreal Ibero	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Space missions are extremely complex projects. They require the development of both the spacecraft and the payload in a coordinated manner and, at the same time, fulfilling many boundary conditions (e.g. budget, time scales, orbit, weight of the mission and others), that often implies the need to look for innovative solutions out-of-the-box that do not appear in textbooks. Moreover, this is usually done by a large team of professionals with an ample variety of backgrounds, skills, and expertise, that need to understand each other and work together for the common objective, even if they do not always speak the same (technical and scientific) language. Future professionals in the aerospace industry need to learn how to work under these conditions, yet this is quite far from what a (science or engineer) student can find in a typical bachelor course. The Mission Assignment in the Space Minor was designed to fill this gap. There, each student team (consisting of about 8 students) will work individually on a Mission Assignment, provided by a Mission Client. The latter can be ESA, SRON, TNO, etc. This Mission Client will provide the mission objectives and top-level requirements. The Space Mission: Scientific Case Study course (hereafter SCS) covers the scientific part of the mission: the payload. Here, the students team translates this into top-level payload design, first. Then, as the course progresses, the parameters of the required observations are determined and fine-tuned. Specifically, instrumental parameters such as, for instance, the central observing wavelength, the total bandwidth of the observations, the spectral resolution, spatial sampling, field of view, or required temporal sampling are considered. These parameters then determine the choice of telescope aperture (to gather enough photons) and detector technology used. In short, the students define the instrumental configuration required to meet the science goals. The course is complemented and should be done together with the twin course Space Mission: Engineering Case Study (code: AE3538), that deals with the spacecraft.	
Study Goals	During the Project Design and Analysis course (see AE3526), each student team worked out an approach on the first scientific analysis and conceptual system design. Based on this, each team provides a preliminary scientific case and key question to be answered, as well as top level requirements for the instrumentation in the spacecraft or space system. The goal of the course is to make the students more familiar with the processes involved in the definition of the scientific aspects of a spatial mission. In particular, the students will find and review the relevant literature on a given science topic. They will need to get a general picture on the science topic and a more detailed insight of the specific question(s) they will address with the mission. Likewise, they will identify similar state-of-the-art missions (or ground-based facilities), and determine and highlight the unique features of this mission. Then, they will establish and investigate the physical relationships between the science target, the optomechanics within the spacecraft, and the resultant detection and processing of the detected signal. In the process, the low level scientific and technical requirements from the initial mission statement and requirements, are defined and refined in an iterative manner.	
Education Method	Each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. The mission, as whole, will start with a kick-off meeting (mandatory attendance). During the course, the students will meet with their team mates and the tutors regularly. Regarding the SCS, the students will have the The Studio Classroom in the Huygens building (Leiden University premises), available one whole day per week for that purpose.	
Assessment	Throughout the course, several review processes will take place. The other teams, the Mission Client and members of the Academic Staff participate in these reviews. More specifically, the global assessment will be the combination of the outcome of oral presentations (mandatory presence), and group and individual reports. Should the student fail to pass any of the items needed for the evaluation, there will be an opportunity for resit in the week after the deadline for submitting the reports (written part), or the week after presentation days (oral part). Details on the format and content for the both the main and resit evaluation procedures will be found on Brightspace.	

CT3533	Earth Observation	3
Responsible Instructor	Dr.ir. J. Timmermans	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Earth observation is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Remote sensing observations from spaceborne instrument inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. This course aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain bio- & geophysical variables iii) to evaluate the different advantages/disadvantages of earth observation technologies. This should allow students to understand how to formulate user requirements for understanding/monitoring Earth into an earth observation mission requirements.	
Study Goals	The overall goal of this course is to provide insight into the physical principles of Earth observation and how earth observation technologies can be used to assess and monitor geo- & biophysical variables on Earth. After this course the student is (cq. you are) able to: LO1: Summarise the signals of the earth in the different parts of the electromagnetic (EM) spectrum and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and subsurface). LO2: Characterise the different observable signals and parameters from Earth and how they can be observed using different remote sensing technologies. LO3: Assess remote sensing technologys advantages/disadvantages for observing geo- & biophysical variables on Earth. LO4: Translate user requirements into mission requirements for earth observation mission. LO5: Evaluate a basic remote sensing analysis to retrieve a geo- & biophysical variable from Earth Observation data.	
Education Method	Classical lectures complemented with pre-recorded videos to maximize interaction in class (flip-the-classroom). Two lab sessions to work on assignment in groups with peer-review.	
Assessment	The students are assessed on their comprehension of the educational material through an assignment. Students are required to perform an assignment where they translate user requirements into mission requirements for a specific earth observation application. Specifically, the students will choose a societal or geophysical topic of interest and area of interest (AOI) to formulate a research question. Select an appropriate satellite mission and get access to the data of this mission Test and select the data analysis platform Analyse the data (this may require some coding) Report your findings in a final report and presentation Through review of the end-of-assignment presentation and final report a grade is given. Because of the nature of the assignment, the assignment can only be performed during the execution of the program.	
Enrolment / Application	This course is intended for students who are registered in the Space Missions minor. If you would like to enroll for this course as elective outside this scopce (e.g. for a 'free minor'), please send an email to course coordinator Joris Timmermans and motivate your request. The course coordinator will decide if you can be admitted.	
Expected prior Knowledge	No specific knowledge required.	
Academic Skills	-	
Literature & Study Materials	Video recordings Book: Rees (2012). Physical Principles of Remote Sensing https://doi.org/10.1017/CBO9781139017411	
Judgement	Students need to score minimal a 6 for the assignment to pass the course. If students do not obtain a passible grade for the assignment the lecturer decides how the deliverable can be amended or remade. For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2023/2024
Organization	Aerospace Engineering
Education	Minors Aerospace Engineering

Programme for EUR students

AE3522UL	Fundamentals of Astronomy	4
Responsible Instructor	M.A. Kenworthy	
Instructor	Dr. M. Rybak	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course provides 2 lectures/instructions per week of 2x45 minutes each. There are three subtopics consisting of Stars, Galaxies and Exoplanets. Each section lasts 2 weeks (with 4 lectures).	
Study Goals	The aim of this course is to give an overview of astronomical and astrophysical phenomena appropriate for the Space Missions Minor. The learning goals are: Understand the origin, evolution and end stages of stars Understand the structure and evolution of galaxies The evolution of the Universe and galaxies over cosmological timescales How we detect extrasolar planets and the search for life in the Universe	
Education Method	The content is provided in lectures throughout the course and with homework sheets every two weeks.	
Literature and Study Materials	Lecture notes available on BrightSpace. Astronomy papers that can be accessed online by the students.	
Prerequisites	This course is only available for students who are registered in the Space Missions minor.	
Assessment	Three assignments (one on each of the topics of Galaxies, Stars and Exoplanets) will be given at the end of that subsection of lectures with a deadline one week after the end of the section. Each will count for 15% of the final grade, for a total of 45% of the final grade. If you fail to submit the homework on time, you get a second opportunity (a resit) to hand the homework in up to one week later, and it will receive only 75% of the score of that specific homework. After one week, the homeworks will be marked with a score of 1. It is not possible to retake a homework based on a low grade. A student needs to compensate with the other homeworks and the exam. The final exam counts for 55% of the final grade. If you fail the course (homeworks + final exam), you can take a retake exam. The best grade of the two exams will count as the final grade for the exam.	

AE3523	Planetary Sciences	3
Responsible Instructor	Dr. S.J. de Vet	
Instructor	Dr. S.M. Cazaux	
Instructor	M.A. Kenworthy	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course Planetary Science will offer you a perspective on the state-of-art in the field of planetary science performed inside the solar system. Past missions and insights from ongoing missions provide a unique perspective on the research performed in this field and new arising science questions. In this course we will reflect on the cross-disciplinary character of the science being performed at the interface of geoscience and astronomy.	
Study Goals	After completion of the course Planetary Science, you will be able to explain how we explore the Solar System and you will apply your scientific knowledge to describe characteristics of planets inside the Solar System and the exoplanets beyond. After this course the student is (cq. you are) able to: LO1: Explain how knowledge on planetary processes and evolution is obtained using planetary exploration missions. LO2: Perform calculations on planetary processes that have been observed inside the Solar System. LO3: Analyse key properties of planets and other planetary bodies inside the Solar System, using data from past and present satellite missions. LO4: Point out the underlying reasons for the physical differences between planets and other planetary bodies inside the Solar System.	
Education Method	Lectures (possibly pre-recorded with online/on-site option for Q&A after streaming of the lecture); two learning labs (i.e., homework assignments), with one dedicated to planetary processes and materials, and one focussed on icy moons, exomoons and habitability.	
Literature and Study Materials	The course and lectures will be based on parts of the highly recommended book: McSween, Jr, H., Moersch, J., Burr, D., Dunne, W., Emery, J., Kah, L., & McCanta, M. (2019). Planetary Geoscience. Cambridge: Cambridge University Press. doi:10.1017/9781316535769 Lecture notes (slides) and assignments will be made available on BrightSpace.	
Assessment	The two learning labs are offered as a homework assignment via BrightSpace and will be graded. Feedback to the student will be provided after grading. A retake for the assignment is possible until 3 weeks after the first submission deadline in the following two cases: (1.) when you miss the deadline (0.2 points will be subtracted per day from the final grade) and (2.) when you fail the assignment you can use the feedback to improve and resubmit the assignment to a maximum grade of 6.0. For all students the final grade for the course will be based on a written exam (60%) and the grades for the learning labs (20% each). The exam will test all theory provided in the lectures slides, the literature (via self-study) and the skills practiced in the provided exercises.	
Enrolment / Application	This course is primarily intended for students who are registered in the Space Missions minor. If you would like to enrol for this course as elective outside this scope (e.g. for a 'free minor'), please send an email to course coordinator dr. Sebastiaan de Vet (s.j.deviet@tudelft.nl) and motivate your request. The course coordinator will decide if you can be admitted.	

AE3524	Spacecraft Design and Astrodynamics	5
Responsible Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr.ir. P.N.A.M. Visser	
Instructor	Dr. J. Guo	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course comprises 20 lectures. Some lectures have more emphasis on theory, some more on exercises. Below is a list of the lectures, the lecturers and its content. Please note that the order of the lectures is indicative and is subject to change. Up-to-date information will be provided on BrightSpace. 1. Spacecraft Basics - Jasper Bouwmeester Brief history of spaceflight Principles and examples of spacecraft Subsystem breakdown & functions 2. Astrodynamics 1/2 - Pieter Visser Brief history orbital mechanics Keplers law Newtons universal law of gravitation Restricted two-Body problem Orbital elements 3. Astrodynamics 2/2 - Pieter Visser Application orbits Ground tracks Orbit perturbations Orbit transfer 4. Structures 1/2 - Ines Uriol Balbin Spacecraft design loads Material selection for space missions Simulation of space structures Testing of space structures 5. Structures 2/2 - Ines Uriol Balbin Structural design Structural analysis 6. Command & Data Handling 1/2 - Alessandra Menicucci Intro to On-Board Data Handling (OBDH) Scope of an on-board computer Data handling system & unit architecture OBDH bus architecture OBDH operations modes 7. Command & Data Handling 2/2 - Alessandra Menicucci Designing an On-Board Computer Radiation effect mitigation Data protection On-Board Computer processor selection Memory selection 8. Electrical Power 1/2 - Jasper Bouwmeester Energy sources Solar power acquisition Voltage regulation Power distribution 9. Electrical Power 2/2 - Jasper Bouwmeester Preliminary sizing Design 10. Thermal Control - Jasper Bouwmeester Thermodynamic basics Passive control Active cooling & heating Sizing and design 11. Attitude Determination & Control 1/2 - Jasper Bouwmeester Orientation & reference frames Sensors overview & working principles Actuators overview & working principles Algorithms: general principles 12. Attitude Determination & Control 2/2 - Jasper Bouwmeester Preliminary sizing Design of sensors Design of actuators	

13. Propulsion 1/2 - Angelo Cervone
Rocket performance equations
Liquid and solid propellant engines
14. Propulsion 2/2 - Angelo Cervone
Electric propulsion
Advanced propulsion systems
15. Communication 1/2 - Stefano Speretta
Basic link budget
Main radio components
Communication chain
16. Communication 2/2 - Stefano Speretta
Link budget error and statistical analysis
Introduction to radio-navigation
17. Guidance, Navigation & Control 1/2 - Jian Guo
Guidance and trajectory planning
Navigation
Kinematics and dynamics
Trajectory control
18. Guidance, Navigation & Control 2/2 - Jian Guo
GNC system architecture design
19. Satellite Design - Guest lecturer
Example(s) of satellite design

Study Goals

The aim of this course is to provide insight in the functions and design options of spacecraft and its astrodynamics. The learning goals are:

Describe spacecraft orbits and calculate its main characteristics.
Formulate the technical principles of spacecraft bus subsystems.
Identify and compare the available subsystem options at component level, calculate their key characteristics and provide their physical and technical limitations.
Characterize the key performance parameters of different spacecraft subsystems and compare the values obtained by ideal theory to the real-life ones.

Education Method

The content is provided in lectures (three per week), of which some contain exercises to be performed, guided and explained in class. Students can self-study by reading annotated hand-outs and performing online exercises. Students will perform one mandatory test for each of the nine topics. Furthermore, AE and astronomy students will perform individual assignment(s) in which they will perform preliminary design of a satellite subsystem or a satellite orbit. This assignment is provided and guided by the associated lectures.

Literature and Study Materials

Annotated lecture slides, available on BrightSpace.
In class exercises. These are part of the lecture slides.
Exercises in ANS for self-study.

Assessment

Recommended book: Wertz and Larson, Space Mission Analysis & Design.

The assessment comprises a (1) digital exam and (2) topic tests for all students. Only for AE and Astronomy students, this is complemented with (3) assignments.

1. Digital Exam

For all students, except AE students, the final grade of AE3524 will be for 100% based on a digital exam in ANS, to be taken in an allocated digital examination room on campus. The exam will test all theory provided in the lectures and skills practiced in the provided exercises and tests. A resit opportunity is provided at the end of the 2nd quarter.

2. Topic Tests

The following setup and rules apply:

- Each student is required to complete tests for each of the 9 main topics of the course.
- Students who have not completed all tests before the start of the exam will be excluded from the exam.
- The tests are taken in ANS and can be completed in multiple sessions.
- Each test is estimated to take between 45 and 90 minutes to complete. This is indicative; there is no time limit applied to the duration of the test.
- Each test will open at the first lecture of the given topic. The lecture will indicate which questions can already be completed after the first lecture.
- A bonus of +0.1 will be applied to the final grade per test which is completed within one week after the concluding (typically second) lecture of each topic with a result of 6.0 or higher.
- Another bonus of +0.1 will be applied to the final grade if all 9 tests are completed in time with sufficient result, which would yield a total bonus of +1.0 in that case.
- No exemptions or deviations from the rules above will be made for unforeseen circumstances, such as illness or other force majeure.
- The questions in the tests are representative for the type and level of questions in the exam. They can be retaken for rehearsal.

3. Assignments (only AE & Astronomy students)

AE students: to account for the partial overlap with the BSC, the examination will count for 60% and one assignment will be provided which counts for 40% of the final grade.
Astronomy students: as alternative to the astronomy course, two assignments will be provided and included in the 4 ECTS extension of this course (AE3524E). Each assignment counts for 50% of its final grade.

The student performs preliminary design of a subsystem or algorithm, including calculations/simulations, according to a set of provided technical requirements. Assignments will be based on one of the nine topics and students can indicate their preferences. Assessment will be based on a small report and a 30-minute discussion with the responsible lecturer on the results. All assignment will be provided in the first lecture week and needs to be completed latest in the final (7th) lecture week. The following rules apply:

- The penalty for exceeding the submission deadline, without prior permission from the supervisor or course coordinator, is a subtraction of 0.1 on the total assignment grade per working day beyond the deadline.
- In case of a force majeure, the supervisor needs to be contacted as soon as possible. A shift of deadline will be provided if this is required and justified.
- When the grade of the assignment is 6.0 or higher, you will not be given the opportunity to improve or do another assignment.
- When the grade is below 5.0, the student must improve the work to be able to pass the course. Between 5.0 and 5.9, the student can choose.
- For potential rework, the supervisor will indicate what needs to be improved. The student then has two more weeks to submit

	the improvement. The assignment grade is the average of the original and the improved work if this leads to a grade of > 6.0, or it becomes the highest grade if it is 6.0 or lower. Only one improvement opportunity will be given. If this is still below 5.0, a new assignment needs to be performed in the next academic year.
Enrolment / Application	This course is only available for students who are registered in the: (1) Space Missions minor (2) AE Space Engineering MSc track who have not followed an Aerospace Engineering BSc nor the Space Missions minor before and want to use this course as elective to get a quick understanding of space engineering. It is not possible to follow this course through other programs (free minor, Erasmus program, master, etcetera).

CT3533	Earth Observation	3
Responsible Instructor	Dr.ir. J. Timmermans	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Earth observation is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Remote sensing observations from spaceborne instrument inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. This course aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain bio- & geophysical variables iii) to evaluate the different advantages/disadvantages of earth observation technologies. This should allow students to understand how to formulate user requirements for understanding/monitoring Earth into an earth observation mission requirements.	
Study Goals	The overall goal of this course is to provide insight into the physical principles of Earth observation and how earth observation technologies can be used to assess and monitor geo- & biophysical variables on Earth. After this course the student is (cq. you are) able to: LO1: Summarise the signals of the earth in the different parts of the electromagnetic (EM) spectrum and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and subsurface). LO2: Characterise the different observable signals and parameters from Earth and how they can be observed using different remote sensing technologies. LO3: Assess remote sensing technologys advantages/disadvantages for observing geo- & biophysical variables on Earth. LO4: Translate user requirements into mission requirements for earth observation mission. LO5: Evaluate a basic remote sensing analysis to retrieve a geo- & biophysical variable from Earth Observation data.	
Education Method	Classical lectures complemented with pre-recorded videos to maximize interaction in class (flip-the-classroom). Two lab sessions to work on assignment in groups with peer-review.	
Assessment	The students are assessed on their comprehension of the educational material through an assignment. Students are required to perform an assignment where they translate user requirements into mission requirements for a specific earth observation application. Specifically, the students will choose a societal or geophysical topic of interest and area of interest (AOI) to formulate a research question. Select an appropriate satellite mission and get access to the data of this mission Test and select the data analysis platform Analyse the data (this may require some coding) Report your findings in a final report and presentation Through review of the end-of-assignment presentation and final report a grade is given. Because of the nature of the assignment, the assignment can only be performed during the execution of the program.	
Enrolment / Application	This course is intended for students who are registered in the Space Missions minor. If you would like to enroll for this course as elective outside this scopce (e.g. for a 'free minor'), please send an email to course coordinator Joris Timmermans and motivate your request. The course coordinator will decide if you can be admitted.	
Expected prior Knowledge	No specific knowledge required.	
Academic Skills	-	
Literature & Study Materials	Video recordings Book: Rees (2012). Physical Principles of Remote Sensing https://doi.org/10.1017/CBO9781139017411	
Judgement	Students need to score minimal a 6 for the assignment to pass the course. If students do not obtain a passible grade for the assignment the lecturer decides how the deliverable can be amended or remade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

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